



DevOps

Other Topics



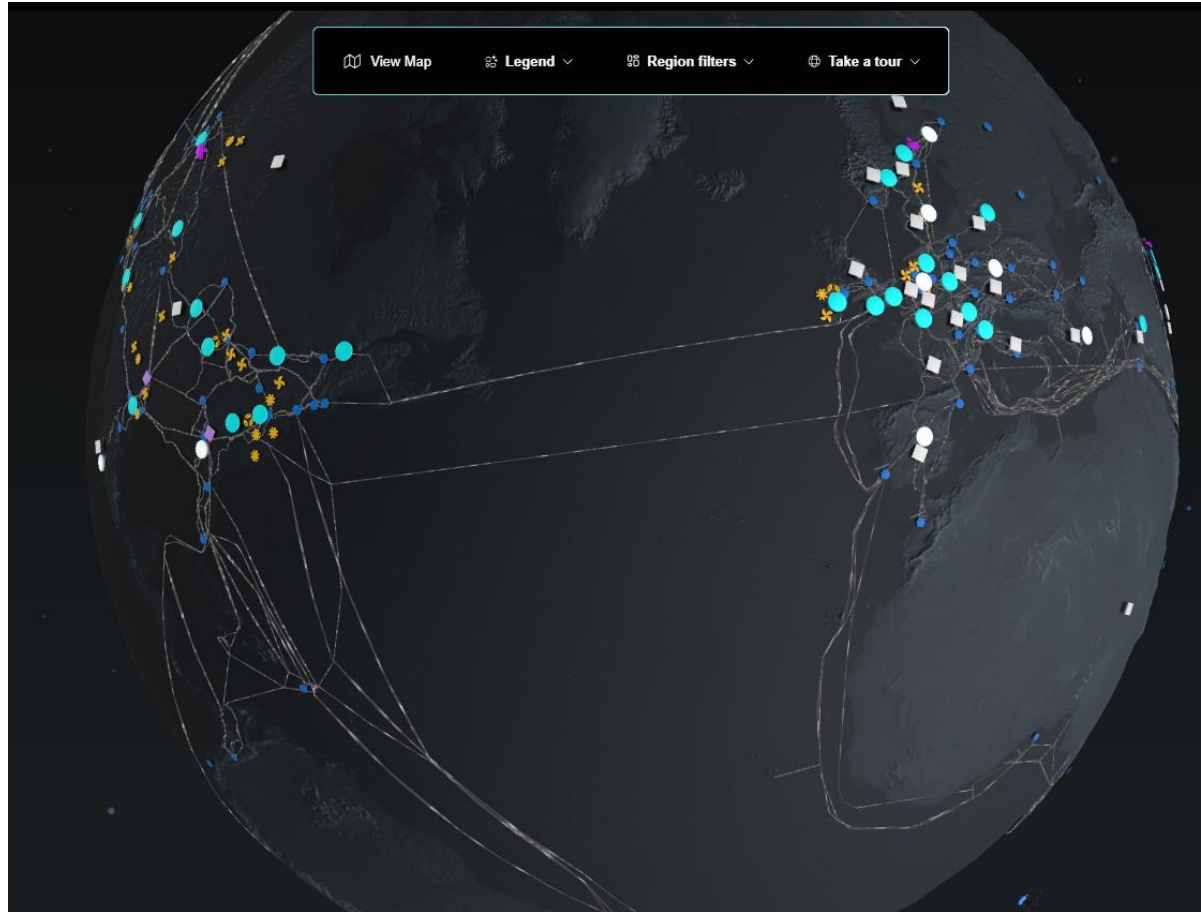
Bibliography

Laszewski, Tom; Arora, Kamal; Farr, Erik; Zonooz, Piyum. Cloud Native Architectures: Design high-availability and cost-effective applications for the cloud (p. 8). Packt Publishing

Scale and Available



Cloud Hyper-Scale



<https://datacenters.microsoft.com/globe/explore>

Hyper-Scale Elements



Multiple Geographic Regions – collection of data centers that operate together



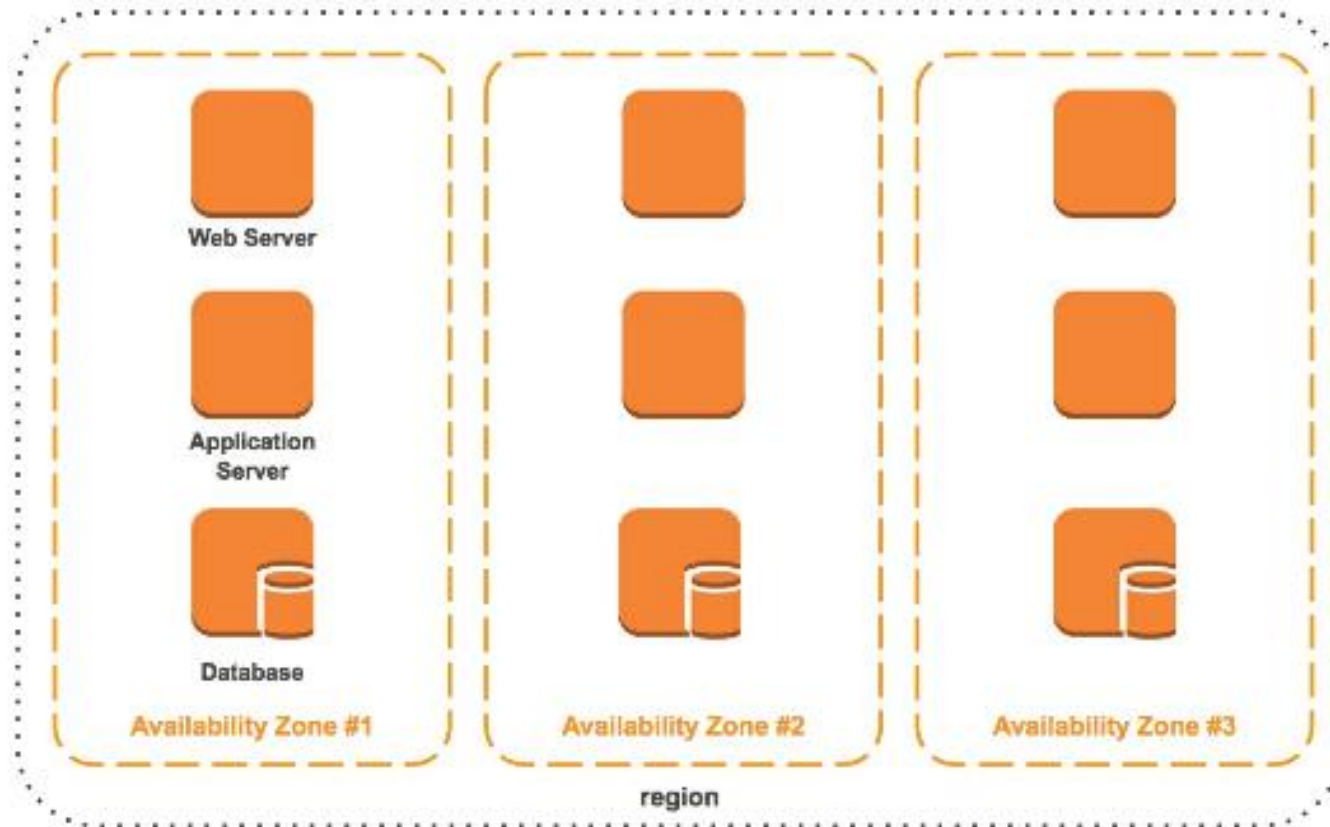
Availability Zones – data center within a GR



High bandwidth trans-oceanic/continental fiber networks to effectively manage traffic between each of data center clusters



Create distributed applications across multiple geographies with very low latency and cost



Applications can be deployed across different AZs – reduce blast radius

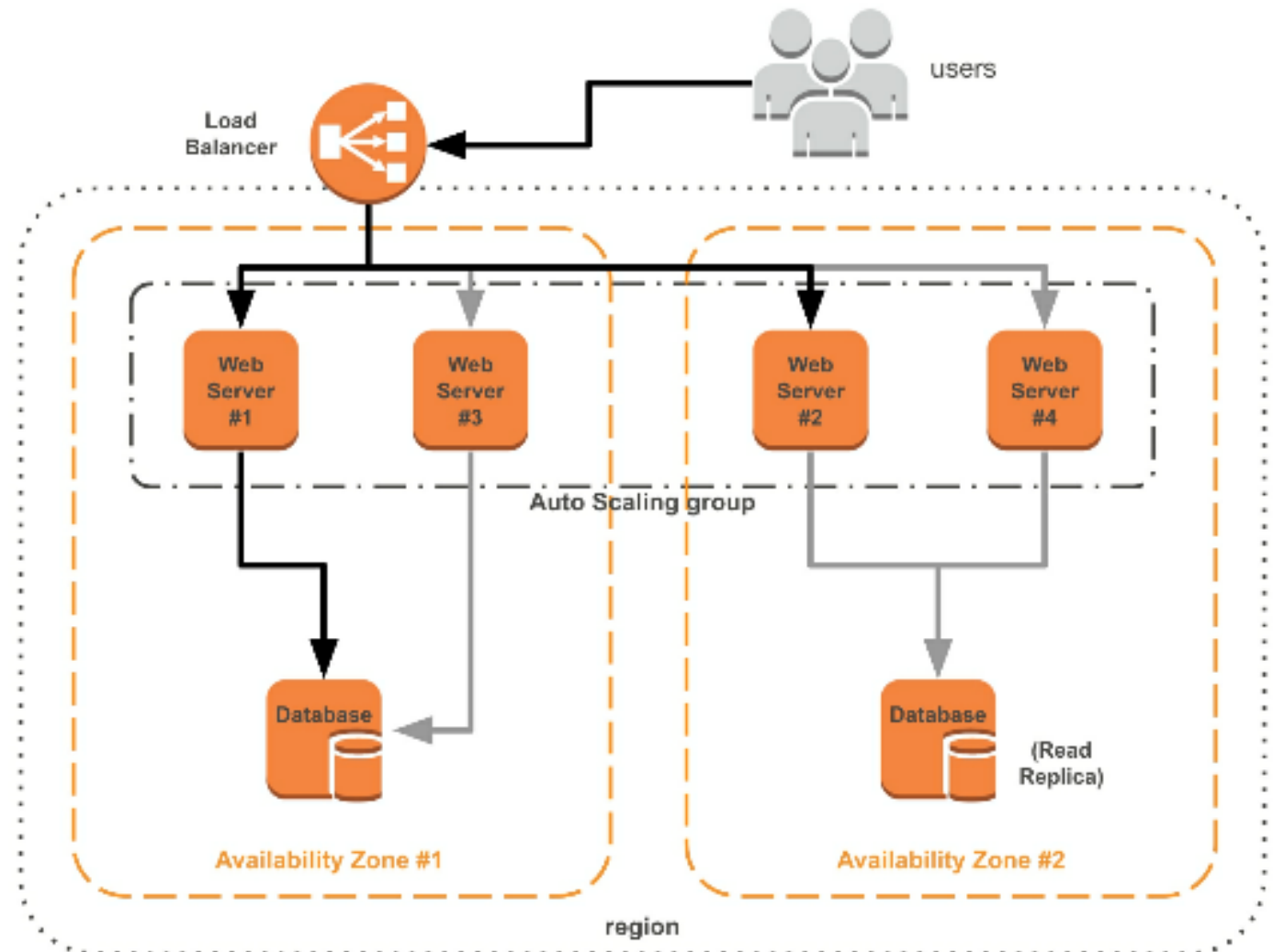
Globally available applications

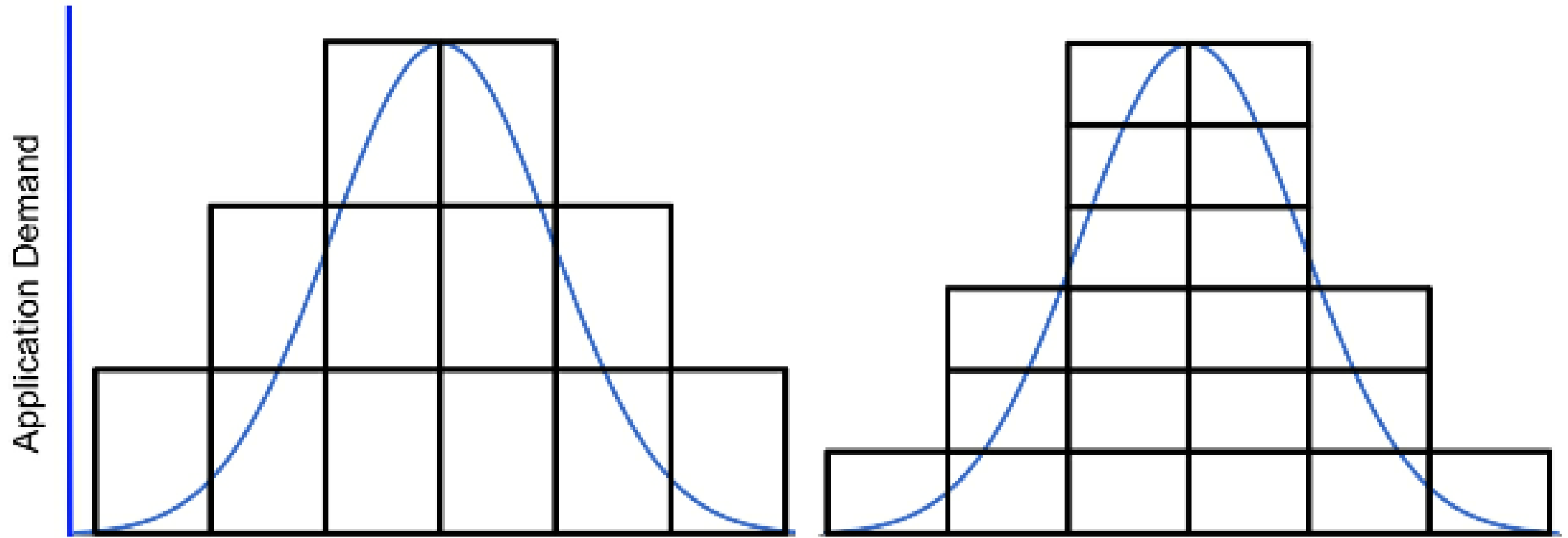
Global DNS service + Load Balancer service

Run across multiple regions

Auto Scaling Groups

Service abstraction that allows users to replicate and expand application VMs dynamically based on various alarms or flags





Smaller VM sizes means for better hardware utilization

Always-on – key architectural
elements

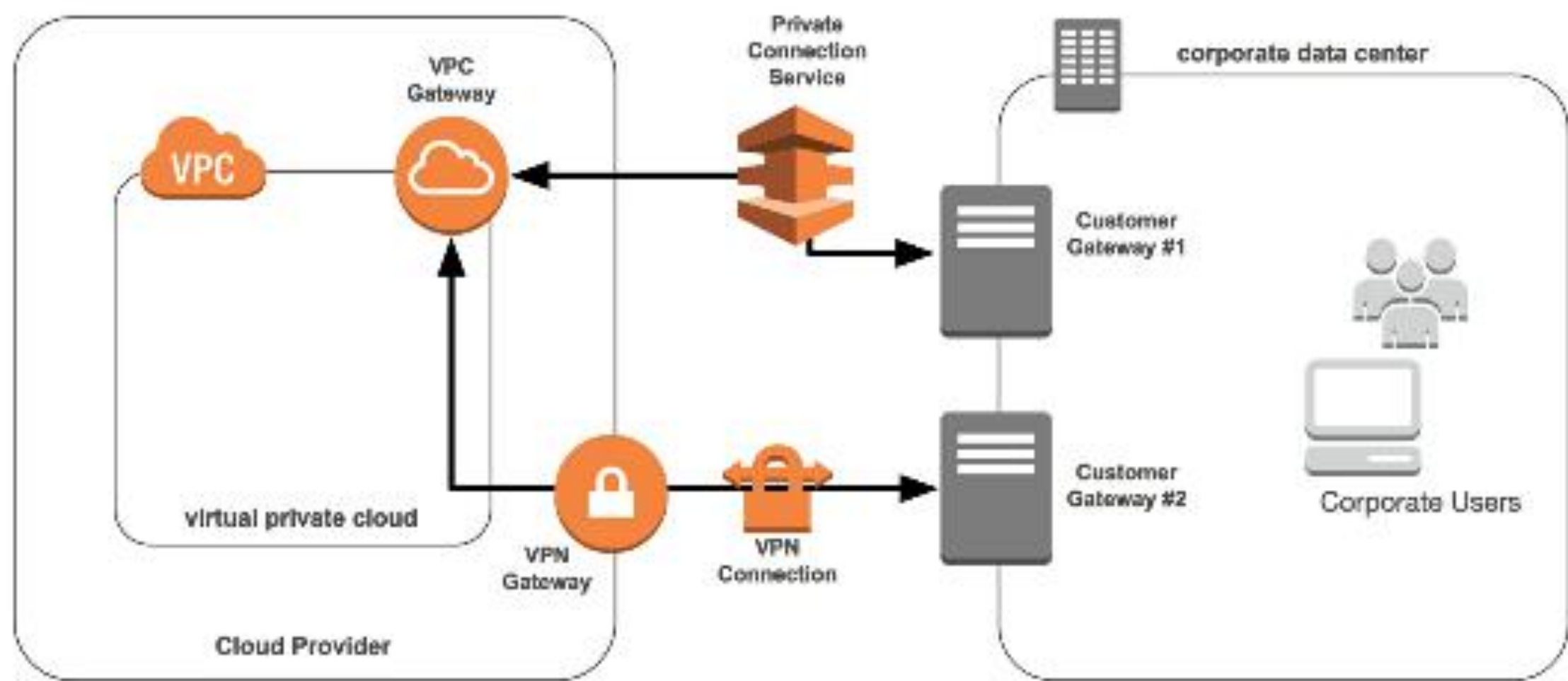
Network Redundancy



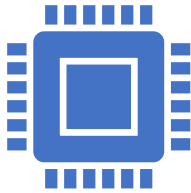
Back up the primary, high bandwidth connection (Express Route) with a failover option (another private connection or a VPN tunnel over the internet) by utilizing separate physical links



Utilize multiple, redundant appliances with parallel network links, mirroring the way cloud providers build out their physical data centers' connectivity



Redundant Core Services



Active Directory (AD)

Rely on connectivity to the on-premise AD server for AD authentication and authorization

Deploy Domain Controller or Read-only Domain Controller in the cloud environment

Offload this undifferentiated heavy lifting to a native cloud service (e.g., Azure Entra)



Domain Name Service (DNS)

Highest level of availability and scalability offered by cloud services (Azure DNS, AWS Route53)



Message based middleware

Deploy queueing system to virtual machines

Use a queue or message bus service offered by the cloud platform (Azure Storage Queue, Azure Service Bus)

Monitoring

If you can't measure it, you can't manage it

Infrastructure Monitoring – collects and reports on the performance of hosts, network, databases, and any other core services that are consumed

Application Monitoring – collects and reports on local resources supporting application runtime, as well as application performance

Real Time User Monitoring – collects and reports all user interactions with the website or client

Log Monitoring – collects and reports all logs that are generated from all hosts, applications, and security devices

Monitoring Options

Native cloud services (Azure Monitor and Azure Log Analytics Workspace, AWS CloudTrail and CloudWatch)

Third-party products (Splunk, Loggly, Dynatrace, Data Dog, New Relic)

A combination of the above

Cloud Native Best Practice: build several dashboards, each giving insights into one application stack or relevant operational groups

Infrastructure as Code &
Immutable Infrastructure
essential for always-on

Self-healing Infrastructures



Inherently smart deployment that is automated to respond to known and common failures



The cloud has completely taken responsibility for "hardware self-healing"

Self-healing at system and application level



Auto-scaling
groups



DNS Health
Checks



Instance (VM)
Autorecovery



Database
failover

Core Principles

Compute is distributed and application is stateless

Storage is non-local and distributed

Create physical redundant network connections

Monitor extensively and exhaustively

Use IaC

Every deployment is immutable

Design and implement self-healing infrastructure whenever possible

Automatically detect and prevent deployment anti-patterns

Create and maintain an Operational Control Plane

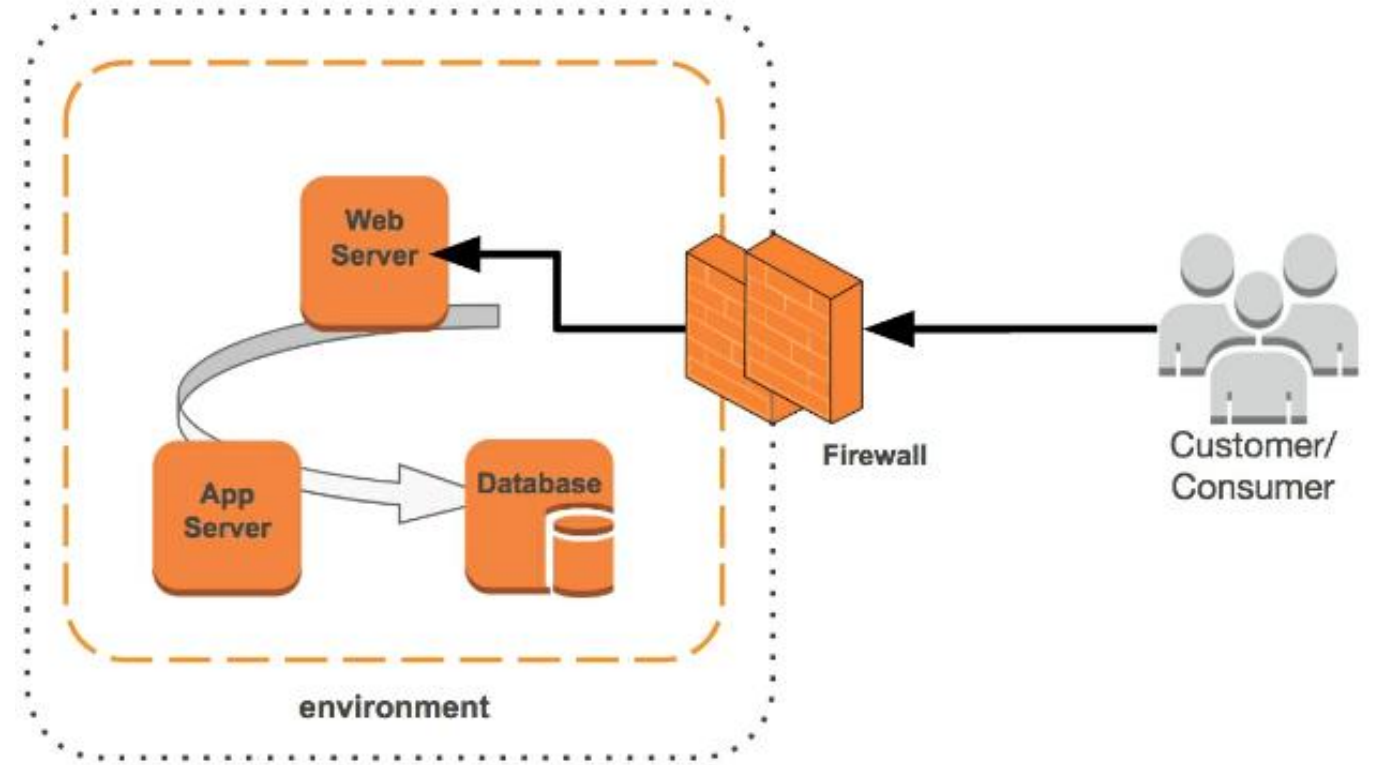
Scalability vs Elasticity

Secure and Reliable



Old model

- Single shell around the application (Firewall)
- Different managers and separate goals from development and operation teams



Security in cloud native world



Provide security products/services in the platform natively



Expose native security features as APIs



Integrate security features into core IT services



Shared responsibility model



Deep security experience offered by cloud provider



Updates and features incorporated into services on a continual basis

Essential features of a secure cloud native stack



Compliant – maintain system compliance and audit trail through automation



Encryption – sensitive data should be encrypted in transit and at rest



Scalable and available encryption resources – do not relay on single encryption resource



DLP - Do not allow any PII or sensitive data to be written to logs or other non-authorized targets



Secure credentials and endpoints – keep service credentials and source/target endpoints outside of the memory



Caching – applications should leverage an external cache to support a stateless design

Cloud Security Services

Network Firewalls – Security Groups

Operates at instance level

Support allow rules only

Return traffic is automatically allowed regardless of any rule

All rules are evaluated before deciding if traffic is allowed

Applies to instance only if SG associated with the instance

Network Firewalls – Network Access Control List

Operates at subnet level

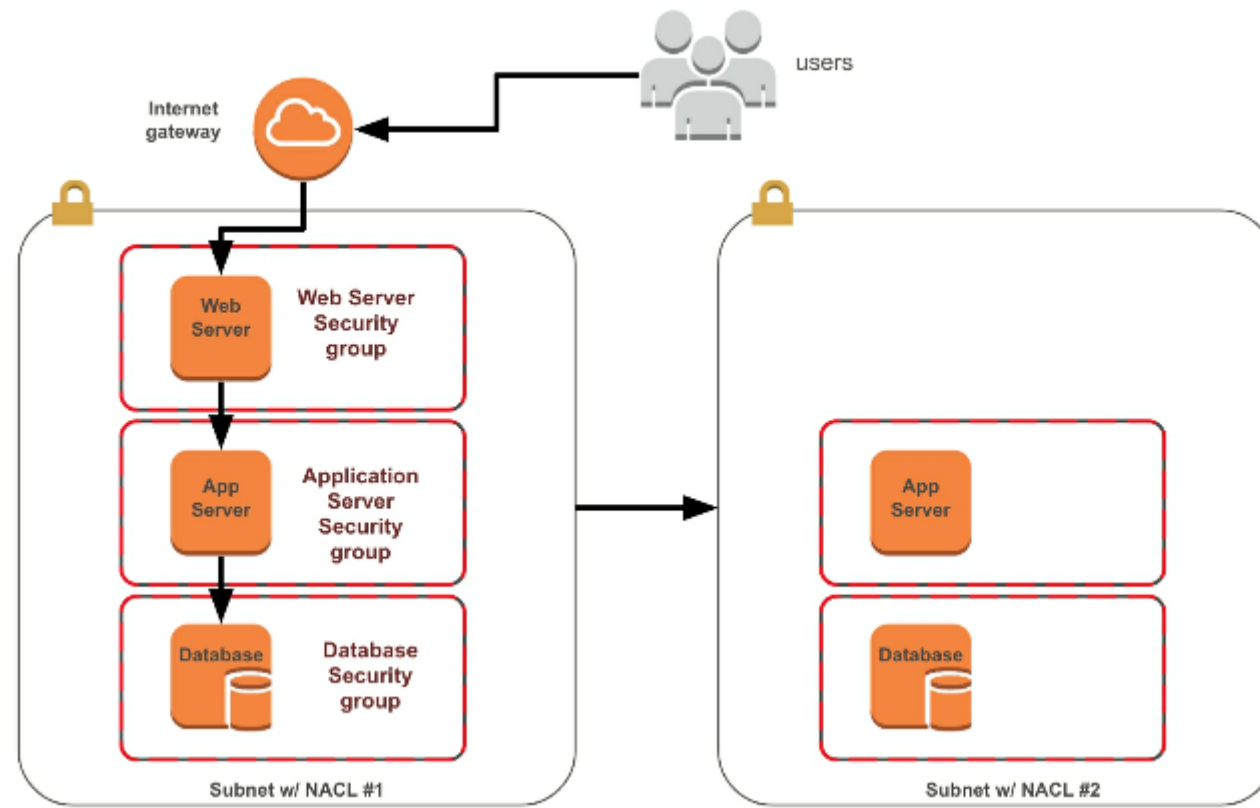
Supports allow and deny rules

Return traffic must be explicitly allowed

Rules are processed in number order when deciding whether to allow traffic

Automatically applies to all instances in the subnet

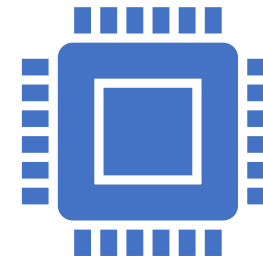
SG vs NACL



Network Logs



Allow the cloud user to view network traffic within their private cloud network plane (Azure Flow logging)



Information available

Source and destination IP addresses

Source and destination ports

IANA protocol number

Number of packets

Size in bytes

Start and end time

Action/Status (allowed / denied)

Network interface ID



Audit logs

- API calls to cloud provider' services
 - Event type, name, and source
 - Access key ID
 - Account ID
 - Username
 - Region
 - Source IP
 - Request parameters
 - Response
-

Monitoring Tools

One pane of glass to have a complete stack monitoring solution has and should always be a goal of security-ops and administrative-ops teams

Detailed monitoring every minute

Set and trigger alarms when monitoring metrics reach certain thresholds

Automate actions based on alarms

Build custom dashboards with graphs and statistics

Configuration Management



Inventory of cloud resources



Inventory of software configuration on the VM



Easy to track any configuration drifts



Identity and access management

- Provision access to users, machines and other services
 - Core Concepts:
 - Users – entity that represents a person or a service (name and secret/access key)
 - Roles – entity that represents a type of access pattern
 - Policies – code statement that implicitly or explicitly grants/rejects access to a cloud service, resource, or object
 - Temporary security credentials – service that grants temporary, limited privilege credentials
 - Cloud providers have developed robust services to authenticate and authorize users and services
-



Encryption services and modules

- Fully managed services that generate and manage cryptographic encryption keys, secrets and certificates (Key Vault)
 - Support for Hardware Security Modules
-



Web application firewalls

- Offered as cloud native services
 - Filter web traffic based on conditions that include IP, HTTP headers, URLs
 - Mitigate common exploits (e.g., SQL injection)
 - Block base on GeoIP
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Optimizing Costs

A solid orange horizontal bar is positioned below the title, spanning a significant portion of the slide's width.

Concerns Before the cloud

Concerns

Physical space

Electric power

Physical security

Network connectivity

Cooling

Physical hardware

Staff

Issues

Large capital investments

Long term project

Cloud cost view

Cloud Service Providers aggregate and reduce costs through economics of scale

What does cloud cost include

- Reduce complexity – eliminates undifferentiated heavy lifting
- Elastic capacity
- Increased speed to market
- Global reach
- Increased operational efficiency
- Enhanced security

Cloud economics



The price of a cloud stack must be compared to the chargeback price plus capital expenditures (building and equipment), operational expenditures (electricity, cooling, water), staffing, licensing (costs for virtualization software, ISV/third-party tools, and so on), facilities, overhead, and opportunity costs

CapEx vs OpEx



Cloud shifts IT spending from Capital Expenditures (CapEx) to Operational Expenditures (OpEx)



Lower, recurring costs versus large, upfront investments



Tax benefits



Greater Transparency



Capital depreciation



Easier growth



No commitment or lock-in

Cost Monitoring



Cloud platforms are built to be cost transparent



Price calculator



Drill down



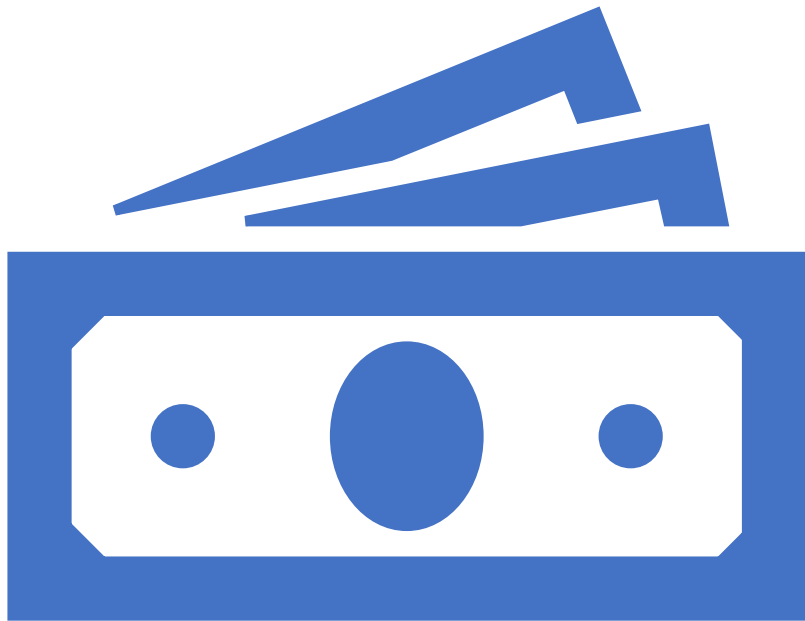
Monitor historical trends



Anticipate future spending



Define budgets



Cost Optimization

Compute Cost Optimization

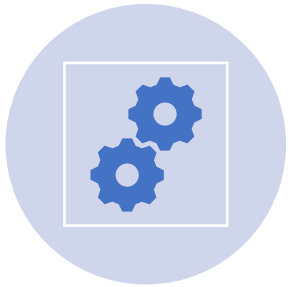


Right sizing = the act of selecting the most adequately sized virtual machine for the given application



Automatically start/stop infrastructure based on needs

Storage Cost Optimization



Establish a life cycle that is as automated as possible



Tags should be used to designate which data stores or objects are needed long-term for compliance purposes



Scripts should be used to manage life cycle



Cloud native services exist to aid users in this effort

Serverless Implications



- Most of the effective cost optimization efforts fall in writing effective code in order to reduce the code's execution time or the number of executions required



Thank you